Printed Pages: 01 **Sub Code: KCA-025**

Paper Id: 232345 **Roll No.** | | | | | | | | | | |

MCA
(SEM III) THEORY EXAMINATION 2022-23
DIGITAL IMAGE PROCESSING

Time: 3 Hours ***Total Marks: 100***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

a. Define neighbors of a pixel.
b. Explain spatial domain processing.
c. What is blind image restoration?
d. Define morphological image processing.
e. What is run length Encoding?
f. What do you mean by image processing?
g. Give various grey level slicing techniques
h. Define the degradation phenomena.
i. Define region growing.
j. What is Arithmetic Coding?

SECTION B

2. Attempt any *three* of the following: 10 x 3 = 30

a. What is meant by image interpolation? Discuss about various interpolation methods.
b. Define Histogram of Image. Explain the concept of Histogram Equalization technique for Image enhancement.
c. Explain about image restoration using inverse filtering. Write the draw backs of this method.
d. How can you control Over segmentation problem? Explain it.
e. Explain about Boundary Extraction and Region Filling Algorithm.

SECTION C

3. Attempt any *one* part of the following: 10 x 1 = 10

a. Illustrate homomorphic filtering approach for image enhancement.
b. What is notch filter? Explain its use in image processing.

4. Attempt any *one* part of the following: 10 x 1 = 10

a. Explain the basic concepts of sampling and quantization in the generation of digital image.
b. Explain about image restoration using minimum mean square error filtering.

5. Attempt any *one* part of the following: 10 x 1 = 10

a. Explain the following two properties of 2D-DFT:
(i) Convolution (ii) Correlation
b. Explain image sharpening using Butterworth high pass and Gaussian high pass filters.

6. Attempt any *one* part of the following: 10 x 1 = 10

a. Explain Huffman coding with an example.
b. Explain about color edge detection with necessary equations.

7. Attempt any *one* part of the following: 10 x 1 = 10

a. Explain edge linking using Hough transform.
b. What are the different image compression standards? Explain.

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